

EchoFind: detecting a submerged body in public sonar data, and what holds up off-site

EchoFind portfolio project · October 2026 · github.com/cheng-huang-ca/echofind

Summary. EchoFind is a handheld-class sonar pipeline that runs from raw echoes to a detector on an edge CPU. It is evaluated the way a rescue crew would meet it: at new sessions, a new frequency and a new site, at a fixed false-alarm rate, against mission time.

- **Detection:** on the UATD forward-looking sonar data, a gradient-boosted model on 38 physics features finds 0.41 of mannequin bodies at a new session, at 0.05 false alarms per frame. A 113k-parameter CNN finds 0.61.
- **Frequency shift:** changing frequency breaks every model. The small CNN recovers the most, but misses the pre-registered keep rule on one of three tests.
- **Calibration and edge:** re-setting the threshold from 50–100 body-free frames restores the false-alarm rate at a new site without any target. The C++ front end matches Python to 3e-7 and processes a 50 m ping in 16 ms at p95 on one core, against a 68 ms budget.
- **Next data:** an 8-day blocked field trial with paired frequencies. Its sample size is set from an intraclass correlation measured on the data.

1. Problem and data

The posting asks for detection that holds across environments, with misses and false alarms both counted. A published CNN on the Bath side-scan data averaged 97% F1 in-distribution but 47% on unseen sites (Nga et al., 2024). EchoFind asks why such gaps appear and what closes them.

Four public sets were used:

- **NOAA HB2305 (EK80 raw pings):** DSP and clutter statistics.
- **MBARI 256 kHz:** real ambient noise.
- **UATD:** Gemini 1200ik, 720 and 1,200 kHz, 9,200 frames, one human-body mannequin among nine confuser classes.
- **Bath side-scan:** 73 mannequin legs at two sites, kept for later.

The Bath archive has lost its target positions, so UATD is the detection set, and Bath has a written labelling protocol (decision 202). UATD logs no site or session; both were inferred from the logged sound speed and range setting (decision 201). Bodies exist only at the lake.

2. Method

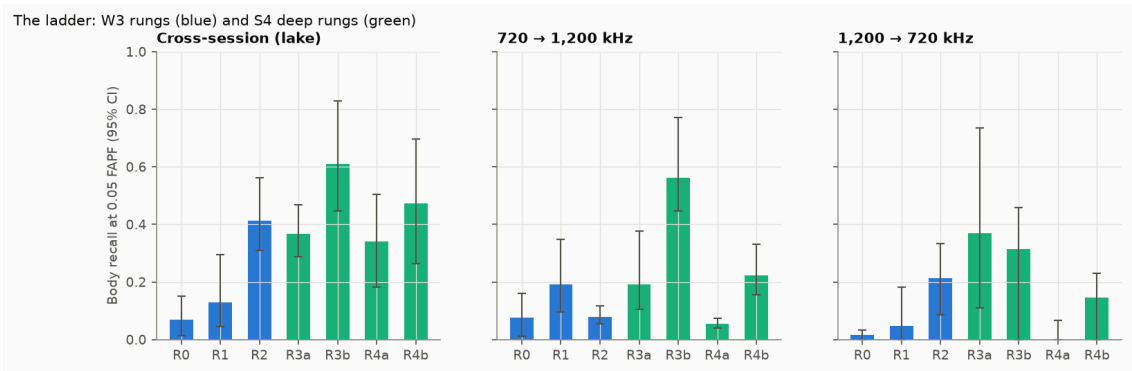
Signal first (W1, W2). NOAA background is heavier-tailed than Rayleigh in 90 of 92 windows (K shape $v \approx 0.5\text{--}0.7$ near the bottom), so fixed thresholds fail. The front end runs IQ demodulation, an FFT matched filter, TVG and CA/GO/OS-CFAR, each checked against closed forms in CI: pulse-compression gain within 0.7 dB of 10 log BT, and Pfa within Monte Carlo error. Near the bottom a body is reverberation-limited, not noise-limited.

A ladder with a keep rule (W3, S4).

- **Shared candidates:** OS-CFAR candidates (23 per frame, 99.7% of bodies proposed) feed every rung.
- **The rungs:** a CFAR + size rule (R0), logistic regression on 10 physics features (R1), LightGBM on 38 features (R2), 1-D and 2-D CNNs (R3a, R3b), and a frozen ResNet-50 and a fine-tuned MobileNetV3 (R4a, R4b).
- **Scoring:** object-level recall at 0.05 false alarms per frame, every non-body candidate counted as an alarm, 95% cluster bootstrap by session.
- **Keep rule:** a rung is kept only if it beats the one below on cross-session and both cross-frequency tests, with paired intervals above zero. The operating point and the rule were committed before any result.

3. Results

Rung	New lake session	720 → 1,200 kHz	1,200 → 720 kHz
R0 rule / R1 logistic	0.07 / 0.13	0.07 / 0.19	0.02 / 0.05
R2 trees, 38 features	0.41 [0.31, 0.56]	0.08 [0.05, 0.12]	0.21 [0.09, 0.33]
R3a 1-D CNN (range profile)	0.37 [0.29, 0.47]	0.19	0.37
R3b 2-D CNN, 113k params	0.61 [0.45, 0.83]	0.56 [0.45, 0.77]	0.31 [0.00, 0.46]
R4a ResNet-50 frozen / R4b MobileNetV3	0.34 / 0.47	0.06 / 0.22	0.00 / 0.15



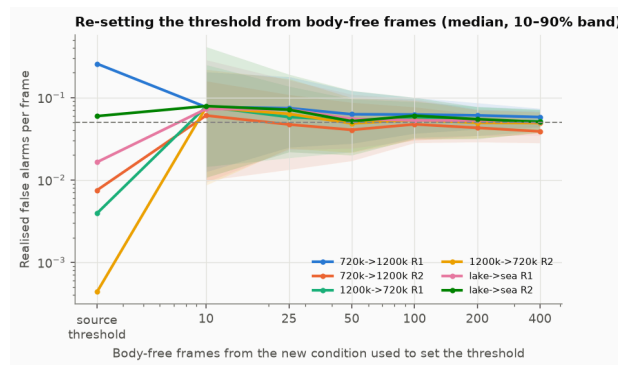
Findings.

1. **Random-frame splits overstate recall 2x:** R2 scores 0.86 on random frames and 0.41 across sessions. Near-duplicate frames sit closer in feature space under the random split.
2. **Each simple rung earns its place across sessions,** and R2 beats R1 by +0.28 [+0.18, +0.39].

3. **Frequency shift breaks every hand-built model.** The small 2-D CNN gains +0.48 over R2 at 720 → 1,200 kHz, but +0.10 [−0.09, +0.25] in the other direction. It misses the rule (decision 602), so R2 remains the shipped model.
4. **Bigger pretrained networks do worse.** The single-beam 1-D CNN only ties R2, so the gain needs a 2-D view.
5. **Shadow features are irreplaceable** (dropping them costs 0.09 recall). Extent features are heavily used but redundant. False alarms come from placed confusers (balls, tyres, cages), not open water.

4. What a crew would see (W4)

- **Search time** = scan time + false alarms × check time + misses × re-search time. Under default times, R2 cuts expected search time from 31 to 23 minutes, while R0 and R1 make it longer. The best false-alarm rate ranges from 0.008 to 0.54 per frame depending on check and re-search costs, so the threshold must come from field numbers.
- **Thresholds drift with conditions:** a source threshold gives 0.0004 to 0.26 false alarms per frame at a new frequency. Re-setting it from 50–100 body-free frames brings it to $0.05 \pm \times 2$ at every shift tested. Body-free frames need no target, so a crew can record them on arrival.
- **Calibration where it matters:** ECE is 0.01 over all candidates but 0.09–0.43 among candidates scored at $p \geq 0.05$.
- **Safety:** under injected faults, input QA catches dropped rows, dead beams and low gain on every frame, but only 56% of saturated frames and no boat wake. An OOD score adds 53% of wake frames. Saturation costs 9 recall points. A model card holds a 10-row hazard table.



5. Edge (W5)

- **Parity:** a C++17 port (zero-phase SOS filtering identical to SciPy, an FFT matched filter, CFAR, TVG) matches Python to $\leq 2e-11$ in double and $\leq 3.1e-7$ in float32. R2 is compiled to C with float thresholds rounded down, so float inputs take the same branches; the float32 cost is −0.002 recall.
- **Latency:** a 200 kHz, 50 m ping takes 16.4 ms at p95 on one pinned x86 core (24% of 68 ms), and 6.1 ms at 80 kHz. R3b would add 8.4 ms.
- **int8:** R3b loses 0.017 recall and shrinks 3.3×, but is not faster on this CPU. Raspberry Pi timing is pending.

6. Limits and what to collect next (W6)

Limits.

- One mannequin, in one lake, in 16 sessions.
- No bodies beyond 12 m.
- The two frequencies were recorded in different sessions.
- No Bath number until its legs are boxed.
- One seed per CNN fold.

Next. The proposed trial runs 4 sites (one per bottom type) × 2 days: 144 body-proxy placements, both frequencies on every ping, a tissue phantom beside a mannequin, confusers, body-free sweeps and timed checks. With 48 placements per arm, it detects a 0.15 recall difference with power 0.89 at the measured ICC (0.12) and 0.52 at a conservative 0.5. A pilot day sets the final size, and a pre-registered sim-to-real test decides whether simulation may serve as training data.